NETFLIX

CONTENT ANALYTICS

Combined Product Design Document + Software Design Document

PROJECT Netflix Dashboard	**PLATFORM** Microsoft Power BI
SOURCE TABLE Netflix Catalog	**STYLE** Dark Cinematic / Red Accent

Prepared from the uploaded Netflix Dashboard Power BI model.

1. PRODUCT DESIGN DOCUMENT (PDD)

1.1 Product Overview

The Netflix Dashboard is an interactive Power BI analytics solution for exploring a streaming-content catalog. The dashboard organizes title volume, movie-versus-TV composition, release-year trends, age-category distribution, audience/rating information, countries, directors and duration-related metrics into multiple analytical views.

1.2 Business Need

- Provide a centralized view of the Netflix catalog.
- Understand the balance between Movies and TV Shows.
- Analyze how title availability changes across release years.
- Explore audience age categories, countries and ratings.
- Identify director coverage and data-quality gaps.
- Summarize movie and TV duration characteristics.

1.3 Product Objectives

- Create an executive-friendly streaming analytics dashboard.
- Convert catalog-level data into measurable KPIs.
- Provide interactive exploration across multiple analytical dimensions.
- Separate executive overview, rating/audience, duration and director/quality analysis.

1.4 Intended Users

User	Primary Use
Content / Catalog Analyst	Explore title mix, release trends, countries and directors.
Business / Management User	Review high-level catalog KPIs and content composition.
Data Analyst	Investigate patterns, distributions and data-quality indicators.
Portfolio Reviewer	Understand the complete Power BI analytical workflow.

2. PRODUCT SCOPE & REQUIREMENTS

2.1 In Scope

Area	Scope
Catalog	Total titles, movies and TV shows.
Content Mix	Percentage split between Movies and TV Shows.
Time	Total titles by release year.
Audience	Title distribution by age category and rating information.
Geography	Distinct countries and title distribution by country.
Duration	Longest/shortest movie, average movie duration and average TV seasons.
Directors	Title distribution by director and missing-director analysis.
Data Quality	Missing director count and missing-director percentage.

2.2 Functional Requirements

ID	Requirement
FR-01	Display total titles, total movies and total TV shows as executive KPIs.
FR-02	Display percentage composition of Movies and TV Shows.
FR-03	Show title trend by release year.
FR-04	Show title distribution by age category.
FR-05	Show mature-content percentage and distinct-country count.

ID	Requirement
FR-06	Show rating availability/distribution by age category.
FR-07	Show title distribution by country.
FR-08	Show movie and TV duration metrics.
FR-09	Show titles by director.
FR-10	Show missing-director count and percentage.

3. ANALYTICAL QUESTIONS

Question	Dashboard Answer
How large is the catalog?	Total Titles KPI.
How is the catalog split?	Movies vs TV Shows KPIs and percentage donut.
How has the catalog evolved?	Release Year trend.
Which age categories dominate?	Age Category distribution.
How broad is geographic coverage?	Distinct Countries KPI and country chart.
What is the rating/audience profile?	Rating distribution by age category.
What are the duration characteristics?	Longest, shortest and average movie duration plus average TV seasons.
How much director information is available?	Director title chart and missing-director KPIs.

4. SOFTWARE DESIGN DOCUMENT (SDD)

4.1 Report Architecture

The uploaded PBIX contains four analytical report pages: **Executive Overview**, **Rating and Audience Insights**, **Duration Analysis**, and **Directors and Quality**. The semantic model contains the main **Netflix Catalog** entity plus a dedicated **Dax measures** table.

Layer	Component	Design Role
Data	Netflix Catalog	Primary catalog dataset used by report visuals.
Measures	Dax measures	Centralized DAX measure layer for reusable KPIs.
Report	Executive Overview	Top-level catalog and content-mix analysis.
Report	Rating and Audience Insights	Audience, rating and geography analysis.
Report	Duration Analysis	Movie/TV duration metrics.
Report	Directors and Quality	Director analysis and data-quality indicators.

4.2 Confirmed Fields Used

The report layout references **Netflix Catalog.Release Year**, **Netflix Catalog.Age_Category**, **Netflix Catalog.Country**, **Netflix Catalog.Director** and **Netflix Catalog.Rating**. These fields are combined with reusable DAX measures from the **Dax measures** table.

5. KPI & DAX MEASURE DESIGN

Measure	Purpose
Total Titles	Overall number of catalog titles.
Total Movies	Number of movie titles.
Total TV Show	Number of TV Show titles.
% Movies	Movie share of the catalog.
% TV Shows	TV Show share of the catalog.
Mature Content %	Mature-content indicator used in audience insights.
Distinct Countries	Geographic coverage indicator.
Longest Movie	Maximum movie duration.
Shortest Movie	Minimum movie duration.
Average Movie Duration	Average movie duration.
Average TV Seasons	Average TV-season metric.
Missing Director	Count of missing director values.
Missing director%	Percentage of records with missing director information.

Design Principle

Measures are separated into the dedicated **Dax measures** table so business calculations can be reused consistently across report pages and visuals.

6. VISUAL & PAGE SPECIFICATION

Page	Confirmed Visuals	Purpose
Executive Overview	3 KPI cards	Total titles, movies and TV shows.
Executive Overview	Donut chart	Movie vs TV Show percentage mix.
Executive Overview	Line chart	Total titles by release year.
Executive Overview	Clustered column chart	Total titles by age category.
Rating and Audience Insights	Card	Mature content % and distinct countries.
Rating and Audience Insights	Donut chart	Rating availability by age category.

Page	Confirmed Visuals	Purpose
Rating and Audience Insights	Clustered bar chart	Total titles by country.
Duration Analysis	Card	Longest/shortest movie, average movie duration and average TV seasons.
Directors and Quality	Bar chart	Total titles by director.
Directors and Quality	Card	Distinct countries, missing directors and missing-director %.

7. UI / UX DESIGN

7.1 Netflix-Inspired Visual Language

- Use a near-black background to create a cinematic streaming-dashboard feel.
- Use red as the primary accent for KPI emphasis, active states and section markers.
- Use white/light-gray typography for strong contrast and readability.
- Use card-based KPI presentation similar to streaming-service executive interfaces.
- Keep analytical pages visually consistent while allowing each page to have a distinct purpose.

7.2 Navigation Concept

- Executive Overview → understand the catalog at a glance.
- Rating and Audience Insights → investigate audience/rating/geographic characteristics.
- Duration Analysis → understand movie and TV duration patterns.
- Directors and Quality → analyze director contribution and missing data.

7.3 Interaction Design

The report uses Power BI visuals and page navigation to separate analytical questions into focused views. Cross-filtering and selection behavior should be validated after refresh to ensure that visual context remains consistent.

8. DATA MODEL & QUALITY DESIGN

8.1 Model Components

Object	Type	Purpose
Netflix Catalog	Data table	Primary content-catalog attributes.
Dax measures	Measure table	Reusable business calculations.

8.2 Data Quality Controls

- Validate title records and categorical fields before refresh.
- Standardize release-year, country, age-category and rating values.
- Monitor missing director records using the dashboard's dedicated measures.
- Check duration fields before calculating averages and min/max values.
- Validate that Movie and TV Show classifications are consistent.

8.3 Privacy Consideration

The dashboard is catalog-focused. Any real-world deployment should still review licensing, source-data permissions and organizational data-governance requirements before publishing or distributing the underlying dataset.

9. TESTING & VALIDATION

Test ID	Test Case	Expected Result
TC-01	Open Executive Overview	All executive KPI and trend visuals render.
TC-02	Validate title KPIs	Total Titles, Movies and TV Shows display correctly.
TC-03	Validate content mix	Movie/TV percentage visual is consistent with KPIs.
TC-04	Validate release-year trend	Years and title totals display without unexpected blanks.
TC-05	Validate age category	Age-category chart displays valid categories.
TC-06	Validate country analysis	Country chart and distinct-country KPI are consistent.
TC-07	Validate rating analysis	Rating/age-category distribution renders correctly.
TC-08	Validate duration metrics	Min/max/average movie and TV-season measures calculate correctly.
TC-09	Validate director analysis	Director chart and missing-director KPIs are populated.

Test ID	Test Case	Expected Result
TC-10	Cross-page navigation	All four report pages are reachable and readable.

Acceptance Criteria

- The four analytical pages communicate distinct business questions without unnecessary duplication.
- KPIs and charts use the intended fields/measures.
- The report remains readable in the selected dark cinematic theme.
- Data-quality measures clearly identify missing director information.
- Documentation matches the implemented PBIX structure.

10. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Usability	Users should understand the purpose of each page quickly.
Performance	Avoid excessive visuals and unnecessarily expensive calculations.
Maintainability	Keep reusable DAX calculations in the dedicated measure table.
Scalability	Allow the catalog to grow without redesigning the entire report.
Data Quality	Validate categorical and duration fields during refresh.
Accessibility	Maintain sufficient text/background contrast despite the dark theme.
Governance	Track source, refresh and distribution permissions.

11. DEPLOYMENT & PORTFOLIO DELIVERY

Deployment Flow

- Validate the Netflix Catalog source data.
- Refresh the Power BI semantic model.
- Validate DAX measures and all four report pages.
- Check navigation, formatting and visual interactions.
- Publish to the intended Power BI workspace when appropriate.
- Apply access controls and data-governance rules.

Recommended GitHub Structure

File / Folder	Purpose
README.md	Project overview, objectives, tools, screenshots and insights.
Netflix Dashboard.pbix	Power BI project file, if sharing is permitted.
Netflix_Dashboard_PDD_SDD.pdf	Combined project documentation.
Screenshots/	Images of the four dashboard pages.
Documentation/	Data dictionary, methodology and supporting notes.

12. FINAL PROJECT SUMMARY

The Netflix Dashboard is a multi-page Power BI streaming-content analytics solution. Its product design focuses on catalog size, content mix, release trends, audience/rating characteristics, geography, duration, directors and data quality. Its software design uses a dedicated Netflix Catalog data entity and a reusable DAX-measure layer across four focused analytical pages.

The document is intentionally based on the uploaded PBIX structure rather than a generic Netflix dashboard concept. The Netflix-inspired styling is used as a presentation theme while the functional and technical descriptions remain aligned with the actual report implementation.

	Netflix Dashboard
	Executive Overview • Rating and Audience Insights • Duration Analysis • Directors and Qual
	Netflix Catalog
	Dax measures
	Microsoft Power BI
	Combined PDD + SDD